Numerical Methods



ODE

Exercise 1: Solve this problem using Matlab/Octave

Problem Statement. Use Euler's method to integrate $y' = 4e^{0.8t} - 0.5y$ from $t = 0$ to 4 with a step size of 1. The initial condition at $t = 0$ is $y = 2$. Note that the exact solution can be determined analytically as

$$y = \frac{4}{1.3}(e^{0.8t} - e^{-0.5t}) + 2e^{-0.5t}$$

Euler's method

Hint

```
dydt=@(t,y) 4*exp(0.8*t) - 0.5*y;

[t,y] = eulode(dydt,[0 4],2,1);

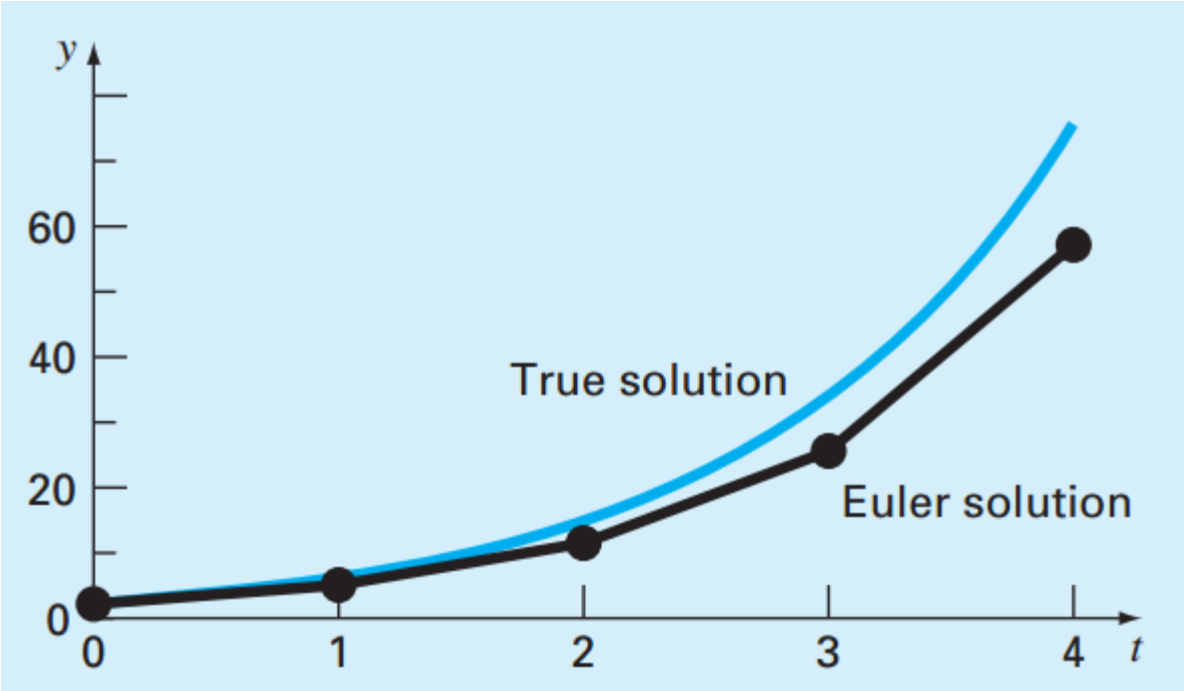
disp([t,y])

plot([t,y])
```

Euler's method

- Matlab

t	y_{true}	y_{Euler}	$\lvert\varepsilon_t\rvert$ (%)
0	2.00000	2.00000	
1	6.19463	5.00000	19.28
2	14.84392	11.40216	23.19
3	33.67717	25.51321	24.24
4	75.33896	56.84931	24.54



this error can be reduced by using a smaller step size.

ODE Solvers in Matlab

Solver	Problem Type	Order of Accuracy	When to use
ode45	Nonstiff	Medium	Most of the time. This should be the first solver tried
ode23	Nonstiff	Low	For problems with crude error tolerances or for solving moderately stiff problems.
ode113	Nonstiff	Low to high	For problems with stringent error tolerances or for solving computationally intensive problems
ode15s	Stiff	Low to medium	If ods45 is slow because the problem is stiff
ode23s	Stiff	Low	If using crude error tolerances to solve stiff systems and the mass matrix is constant
ode23t	Moderately stiff	Low	For moderately stiff problems is you need a solution without numerical damping
ode23tb	Stiff	Low	If using crude error tolerances to solve stiff systems

Exercise 2

Solve the ODE using Matlab/Octave:

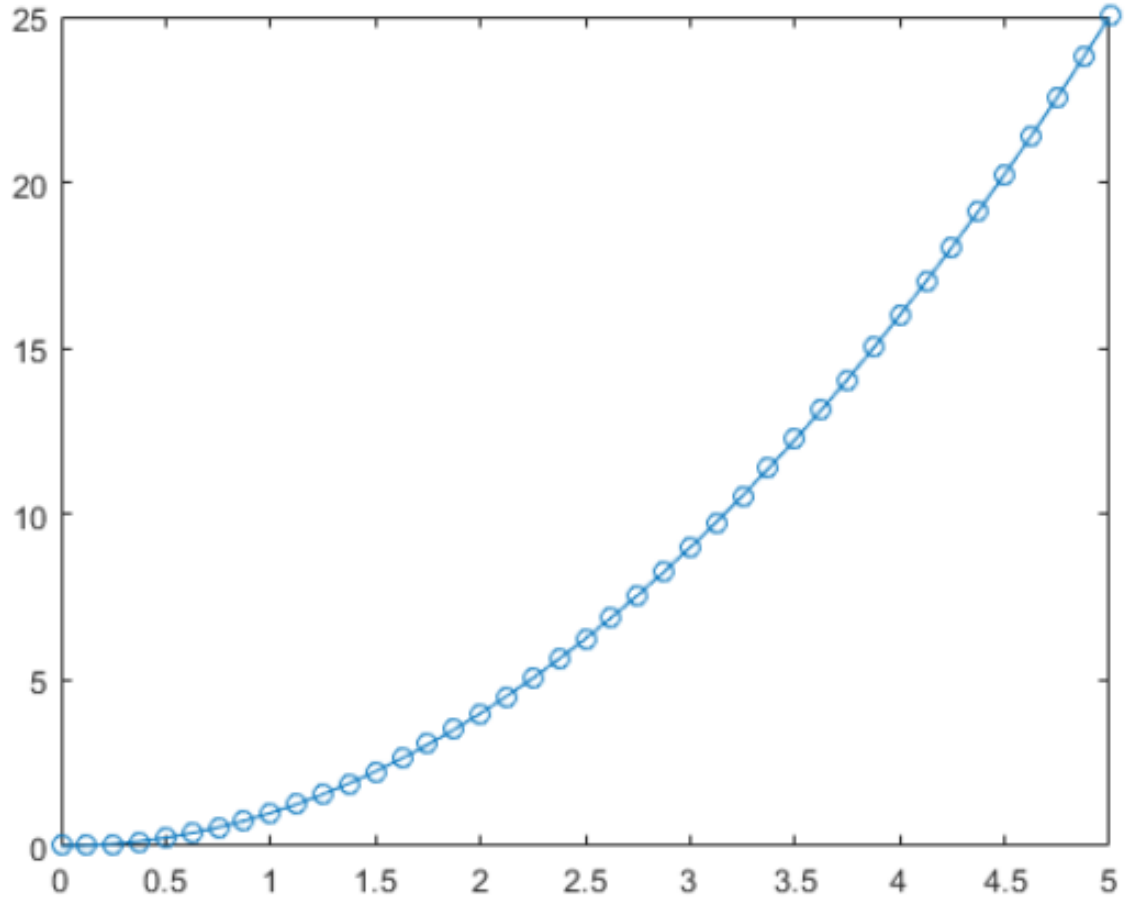
$$y' = 2t$$

Use a time interval of [0,5] and the initial condition y0 = 0.

Hint:

```
tspan = [0 5];
y0 = 0;
[t,y] = ode45(@(t,y) 2*t, tspan, y0);

plot(t,y,'-o')
```



Fourth-Order Runge-Kutta Methods

$$y_{i+1} = y_i + \phi(x_i, y_i, h)h \qquad \phi = a_1k_1 + a_2k_2 + \cdots + a_nk_n$$

The fourth-order version

$$y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h$$

where

$$k_1 = f(x_i, y_i)$$

$$k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1h\right)$$

$$k_3 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_2h\right)$$

$$k_4 = f(x_i + h, y_i + k_3h)$$

Exercise 3: using Matlab/Octave

$$\begin{cases} y' = y - t^2 + 1 \\ y(0) = 0.5 \end{cases}$$

The exact solution for this problem is $y = t^2 + 2t + 1 - \frac{1}{2}e^t$, and we are interested in the value of y for $0 \leq t \leq 2$.

Formula:

$$y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h$$

where

$$k_1 = f(x_i, y_i)$$

$$k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h\right)$$

$$k_3 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_2 h\right)$$

$$k_4 = f(x_i + h, y_i + k_3 h)$$

Analytical solution

$$\begin{cases} y' = y - t^2 + 1 \\ y(0) = 0.5 \end{cases} \qquad 0 \le t \le 2.$$

Now let's compare what we got with the exact solution

t_i	Exact solution $y(t_i)$	Numerical solution w_i	Error $\lvert w_i - y(t_i)\rvert$
0.0	0.5	0.5	0
0.5	1.425639364649936	1.425130208333333	0.000509156316603
1.0	2.640859085770477	2.639602661132812	0.001256424637665
1.5	4.009155464830968	4.006818970044454	0.002336494786515
2.0	5.305471950534675	5.301605229265987	0.003866721268688

Runge-Kutta methods

Hint:

$$\begin{cases} y' = y - t^2 + 1 \\ y(0) = 0.5 \end{cases} \qquad 0 \le t \le 2.$$

```
function rungekutta
  h = 0.5;
  t = 0;
  w = 0.5;
  fprintf('Step 0: t = %12.8f, w = %12.8f\n', t, w);
  for i=1:4
    k1 = h*f(t,w);
    k2 = h*f(t+h/2, w+k1/2);
    k3 = h*f(t+h/2, w+k2/2);
    k4 = h*f(t+h, w+k3);
    w = w + (k1+2*k2+2*k3+k4)/6;
    t = t + h;
    fprintf('Step %d: t = %6.4f, w = %18.15f\n', i, t, w);
  end
```

```
function v = f(t,y)
  v = y-t^2+1;
```

Runge-Kutta methods

Continue with h=0.2

$$\begin{cases} y' = y - t^2 + 1 \\ y(0) = 0.5 \end{cases} \qquad 0 \le t \le 2.$$

2. Solve the problem using RK4 with $h = 0.2$. All you need to do is to replace `h = 0.5;` and `for i=1:4` in the above Matlab program into `h = 0.2` and `for i=1:10`. Then we have

t_i	Exact solution $y(t_i)$	Numerical solution w_i	Error $\lvert w_i - y(t_i)\rvert$
0.0	0.5	0.5	0
0.2	0.829298620919915	0.829293333333333	0.000005287586582
0.4	1.214087651179365	1.214076210666667	0.000011440512698
0.6	1.648940599804746	1.648922017041600	0.000018582763146
0.8	2.127229535753766	2.127202684947944	0.000026850805823
1.0	2.640859085770477	2.640822692728752	0.000036393041726
1.2	3.179941538631726	3.179894170232231	0.000047368399496
1.4	3.732400016577663	3.732340072854980	0.000059943722683
1.6	4.283483787802442	4.283409498318406	0.000074289484036
1.8	4.815176267793527	4.815085694579435	0.000090573214092
2.0	5.305471950534674	5.305363000692655	0.000108949842019

Continue with h=0.05

$$\begin{cases} y' = y - t^2 + 1 \\ y(0) = 0.5 \end{cases} \qquad 0 \le t \le 2.$$

3. Solve the problem using RK4 with $h = 0.05$**.** Again, all need to do is to set `h = 0.05` and `for i=1:40`. Then we have

t_i	Exact solution $y(t_i)$	Numerical solution w_i	Error $\lvert w_i - y(t_i)\rvert$
0.0	0.5	0.5	0
0.05	0.576864451811988	0.576864446614583	0.000000005197405
0.10	0.657414540962176	0.657414530368210	0.000000010593966
...	...	...	...
1.80	4.815176267793529	4.815175898599096	0.000000369194433
1.85	4.942590238699086	4.942589852008494	0.000000386690592
1.90	5.067052778860367	5.067052374183828	0.000000404676539
1.95	5.188156209705356	5.188155786548850	0.000000423156505
2.00	5.305471950534677	5.305471508400809	0.000000442133868

$$\begin{cases} y' = y - t^2 + 1 \\ y(0) = 0.5 \end{cases} \qquad 0 \le t \le 2.$$

4. Comparing the results By comparing the above results, we can see that

$$\begin{aligned} &\text{for } h = 0.5, \quad \text{the error at } t = 2 \text{ is } 0.003866721268688, \text{ using } 4 \text{ steps} \\ &\text{for } h = 0.2, \quad \text{the error at } t = 2 \text{ is } 0.000108949842019, \text{ using } 10 \text{ steps} \\ &\text{for } h = 0.05, \quad \text{the error at } t = 2 \text{ is } 0.000000442133868, \text{ using } 40 \text{ steps} \end{aligned}$$

This brings out a question. How to find the most appropriate step size, if we want to make sure the error is less than a given ε, for example, $\varepsilon = 0.00001$?

Runge-Kutta methods

Full code

$$\begin{cases} y' = y - t^2 + 1 \\ y(0) = 0.5 \end{cases} \qquad 0 \le t \le 2.$$

```
h=0.05;                                          % step size
x = 0:h:100;                                      % Calculates upto y(3)
y = zeros(1,length(x));
%y(1) = [-0.5;0.3;0.2];
y(1) = -0.5; % redo with other choices here.
    % initial condition
F_xy = @(t,r) 3.*exp(-t)-0.4*r;                  % change the function as you desire
for i=1:(length(x)-1)                            % calculation loop
    k_1 = F_xy(x(i),y(i));
    k_2 = F_xy(x(i)+0.5*h,y(i)+0.5*h*k_1);
    k_3 = F_xy((x(i)+0.5*h),(y(i)+0.5*h*k_2));
    k_4 = F_xy((x(i)+h),(y(i)+k_3*h));
    y(i+1) = y(i) + (1/6)*(k_1+2*k_2+2*k_3+k_4)*h;  % main equation

end
```

```
% validate using a decent ODE integrator
tspan = [0,100]; y0 = y(1);
[tx, yx] = ode45(F_xy, tspan, y0);

plot(x,y,'o-', tx, yx, '--')
```

Exercise 4: RF4 in Matlab/Octave

Similarly, integrate

$f(x, y) = 4e^{0.8x} - 0.5y$

using $h = 0.5$ with $y(0) = 2$ from $x = 0$ to 0.5.

the true solution of 3.751521.

Check: Analytical solution and RF4 solution

Similarly, integrate

$f(x, y) = 4e^{0.8x} - 0.5y$

using $h = 0.5$ with $y(0) = 2$ from $x = 0$ to 0.5.

the true solution of 3.751521.

$$\phi = \frac{1}{6}[3 + 2(3.510611) + 2(3.446785) + 4.105603] = 3.503399$$

$$y(0.5) = 2 + 3.503399(0.5) = 3.751699$$

Bonus Exercise

Example 1. Let $y(t)$ be the solution of the initial-value problem

$$dy/dt = 1 + (y - t)^2, \qquad y(0) = \tfrac{1}{2}.$$

Use the fourth-order Runge-Kutta methods with step sizes of 0.1 and 0.5 to solve for y between t = 0 and 1.

The exact solution of this initial-value problem (see Exercise 7) is

$$y(t) = t + 1/(2 - t).$$

Bonus Exercise

26.1 Given

$$\frac{dy}{dt} = -100{,}000y + 99{,}999e^{-t}$$

(a) Estimate the step-size required to maintain stability using the explicit Euler method.

(b) If $y(0) = 0$, use the implicit Euler to obtain a solution from $t = 0$ to 2 using a step size of 0.1.